

Answers

1. Is addition associative/commutative? Show an example.

Answer: Both

Commutative: $1 + 2 = 2 + 1$

Associative: $1 + (2 + 3) = (1 + 2) + 3$

2. Subtraction? Show an example.

Answer: Neither

Commutative: $1 - 2 = -1$

$2 - 1 = 1$

Associative: $1 - (2 - 3) = 2$

$(1 - 2) - 3 = -4$

3. Multiplication? Show an example.

Answer: Both

Commutative: $1 \times 2 = 2 \times 1$

Associative: $1 \times (2 \times 3) = (1 \times 2) \times 3$

4. Division? Show an example.

Answer: Neither

Commutative: $1 \div 2 = \frac{1}{2}$

$2 \div 1 = 2$

Associative: $1 \div (2 \div 3) = \frac{3}{2}$

$(1 \div 2) \div 3 = \frac{1}{6}$